# LASTNAME_16

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**Follow The Below Instructions: .**

**1.The exam duration is 2 hours.**

**2.Read all questions carefully before answering.**

**3.Answer all required questions.**

Examiner Part Only:

**Q1:** What is the difference between population and sample? [2]

**Q2:** Explain mean, median, and mode. [3]

**Q3:** What is standard deviation, and why is it important? [3]

**Q4:** What is the Central Limit Theorem? [2]

**Q5:** Explain Type I and Type II errors. [3]

**Q6:** What are window functions? [3]

**Q7:** What is the difference between correlation and causation? [3]

**Q8:** What are common probability distributions (Normal, Binomial, Poisson)? [3]

**Q9:** When would you use a t-test instead of a z-test? [3]

**Q10:** Explain conditional probability. [4]

A AA

B BB

C CC

D DD

**Q11:** What is the expected value of a random variable? [3]